

AMAL GEORGE

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PROFESSIONAL SUMMARY

Cybersecurity-focused Full Stack and Web3 Developer specializing in secure application design, smart contract auditing, end-to-end encryption, and MITRE ATTACK-based threat detection. Engineered a SOC simulation platform with rule-based intrusion detection, an E2EE chat system using hybrid RSA/AES cryptography, and a decentralized Web3 dApp leveraging Solidity and Hardhat with custom on-chain governance. Combines production MERN development and EVM smart contract development with hands-on penetration testing (Kali Linux, Nmap, Metasploit).

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Frontend: HTML, CSS, Javascript, React, Web3

Backend: Node.js, Express.js, REST APIs, JWT, MVC

Databases: MongoDB, MySQL

Security: Penetration Testing, Vulnerability Assessment, End-to-End Encryption, MITRE ATTACK Mapping

Developer Tools: VS Code, Git, GitHub, Postman, AWS, Solidity

EDUCATION

IIIT Kottayam

BTech in Computer Science with Specialization in Cyber Security

2023 – 2027

CGPA : 7.03

Vijayagiri Public School

Higher Secondary Education

2020 – 2022

Grade : 91.6

WORK EXPERIENCE

Full Stack MERN Developer

Jul 2025 – Jan 2026

Labmentix Private Limited

- Developed and maintained full-stack web applications using the MERN stack, delivering 10+ features and modules to improve application functionality, scalability, and user experience.
- Designed and integrated 20+ RESTful APIs, implemented JWT-based authentication, and optimized MongoDB queries, achieving a 10% improvement in API response time.
- Built responsive React.js interfaces, resolved 15+ production bugs and led code reviews across a Git-based team workflow, improving application stability and reducing regression incidents

PROJECTS

CRYPTCHAT | MERN Stack, Socket.IO, MongoDB, JWT, RSA-OAEP, AES-256-GCM

May 2026

- Built a security-focused real-time chat application using the MERN stack with Socket.IO, featuring End-to-End Encryption (E2EE) for confidential messaging.
- Executed hybrid encryption using RSA-OAEP, AES-256-GCM, and PBKDF2 to securely exchange keys, encrypt messages, and protect user's private keys on the client side.
- Integrated JWT authentication, HTTP-only cookies, Helmet, rate limiting, encrypted message storage, online user tracking, live notifications, and persistent chat history.

THREATLENS | Python, React, FastAPI, Node.js, MongoDB, JWT

June 2026

- Designed and developed a SOC simulation platform that parses Linux authentication logs and correlates security events to identify malicious authentication behavior.
- Engineered a rule-based detection engine capable of identifying brute-force attacks, root account targeting, username enumeration, and distributed authentication attacks with MITRE ATTACK mapping and risk scoring.

- Implemented secure authentication, interactive dashboards, alert management, and historical analysis using React, FastAPI, Express, MongoDB, and Chart.js.

FUNDCHAIN | *Blockchain, React, Solidity, Node.js, Hardhat, Sepolia, Metamask*

September 2026

- Architected and deployed a decentralized Web3 crowdfunding dApp on the Sepolia testnet using Solidity, Hardhat, and Ethers.js v6, implementing a factory pattern to dynamically spin up isolated smart contracts.
- Engineered a trustless fund-governance model utilizing custom smart contract logic to lock raised capital and enforce majority-voting among contributors before executing spending proposals.
- Developed a responsive, dark-mode Web3 dashboard using React.js and Tailwind CSS, enabling real-time transaction state updates, wallet connection via MetaMask, and intuitive contract interactions.

CERTIFICATIONS

MERN Stack Web Development

Nov 2024 – Jun 2025

Udemy – MasynTech

- Developed full-stack web applications using React, Node.js, Express.js, and MongoDB.
- Implemented RESTful APIs, JWT authentication, and secure database schemas across production-style projects.

Ethical Hacking & Security

Sept 2025 – Dec 2025

Udemy – zSecurity

- Executed penetration testing, vulnerability assessments, and web/network security analysis.
- Utilized Kali Linux, Nmap, Wireshark, and Metasploit to identify and exploit security vulnerabilities.

Ethereum & Solidity Web3 Development

July 2026 – Aug 2026

Udemy – Stephen Grider

- Engineered and deployed decentralized applications (dApps) using Solidity, Hardhat, Ethers.js, and React.
- Designed smart contract architectures implementing factory patterns, access controls, and custom governance logic.